



Uniform Scattering of Robots on Alternate Nodes of a Grid

Group No.: 20

Date: 17th December 2025

Mentor By:

Sruti Gan Chaudhuri

Presented By:

Sudip Naskar

Roll No: 002211001131

Department: Information Technology

Subhadip Sinha

Roll No: 102111001154

Department: Information Technology

Subhas Chandra Shit

Roll No: 002211001130

Department: Information Technology

Table of Contents



01 Introduction

02 Motivation & Background

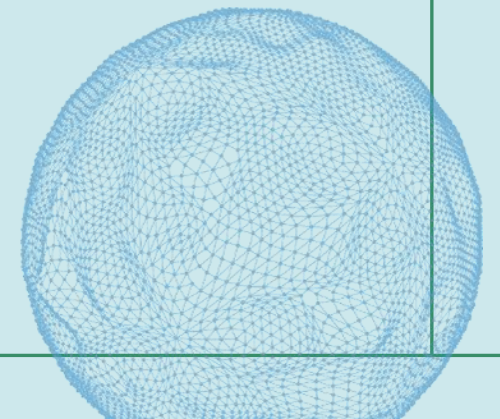
03 Problem Definition

04 Robot Model

05 System Framework

06 Algorithm Overview

07 Output

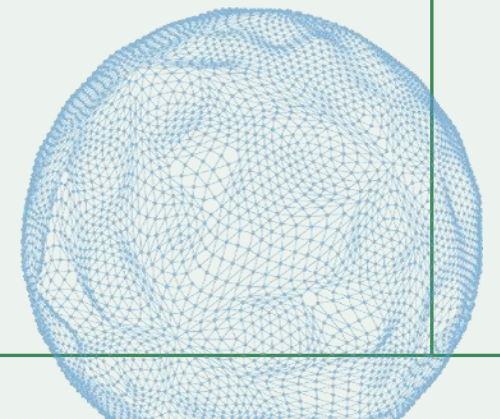
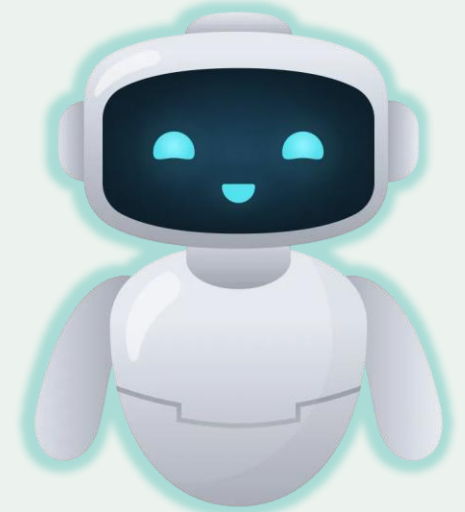


INTRODUCTION

Swarm robotics is an area of distributed computing and robotics that studies the collective behavior of multiple autonomous robots operating in a shared environment. In such systems, robots are typically simple, identical, and operate without centralized control or explicit communication.

Each robot follows a repeated cycle of sensing its surroundings, computing a decision based on local observations, and executing movement. Despite these limitations, swarm robotic systems are capable of exhibiting coordinated and organized global behavior through local interactions alone.

This seminar discusses a distributed approach for organizing autonomous mobile robots on a two-dimensional grid under a semi-synchronous execution model, highlighting how simple local rules can lead to structured and predictable global configurations.



MOTIVATION & BACKGROUND

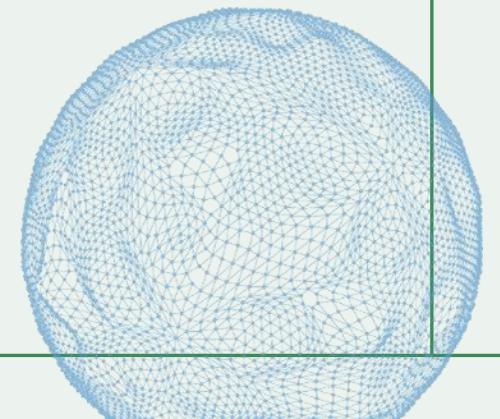
Uniform distribution of robots is essential for effective area coverage.

Traditional deployment methods often rely on manual placement or fixed sensors.

Such approaches usually provide uneven and incomplete coverage in large spaces.

Robots may cluster in some regions while other areas remain uncovered.

This reduces monitoring efficiency and increases the risk of missed regions.



PROBLEM DEFINITION

Robots are initially placed at arbitrary positions on a two-dimensional grid.

There is no guarantee that the robots are evenly spaced or well organized.

The robots are autonomous, anonymous, and have no memory of past actions.



They cannot communicate with each other and rely only on observation.

The objective is to uniformly scatter the robots on alternate nodes of the grid.

The solution must avoid collisions and deadlock and converge in finite time.

ROBOT MODEL & ASSUMPTIONS

Execution Model

- ✓ Robots follow the Look-Compute-Move cycle
 - **Look:** observe positions of all robots on the grid
 - **Compute:** decide the next move based on the observation
 - **Move:** execute the computed movement
- ✓ The system follows a semi-synchronous (SSYNC) model, where a subset of robots is activated simultaneously in each round

Robot Properties

- ✓ Robots operate autonomously without the need for a central controller.
- ✓ They are anonymous and identical in behavior and capability
- ✓ The system is oblivious, meaning robots do not retain memory of past actions.
- ✓ Each robot has unlimited visibility of the entire grid, while movement is restricted exclusively to the edges.

SYSTEM FRAMEWORK

Environment

- ✓ Infinite two-dimensional grid
- ✓ Robots initially placed at distinct nodes
- ✓ Grid nodes are uniformly spaced
- ✓ Robots occupy only grid nodes at any time

Agreements

- ✓ Common X-axis and Y-axis directions
- ✓ No shared global origin
- ✓ No unique robot identifiers
- ✓ No direct communication between robots



KEY IDEA OF THE ALGORITHM



High-Level Strategy

- ✓ Compute the required grid dimensions
- ✓ Identify the north-most and west-most boundaries
- ✓ Gradually reposition robots to:
 - Alternate rows
 - Alternate columns
- ✓ Achieve uniform spacing without collisions

MAIN ALGORITHM

Algorithm 3: FormGrid(R) — Part I

Input : Set of n robots R
Output: Robots move to form a uniform grid distribution
Calculate $rc, d \leftarrow \text{FindDimension}(n)$;
 rc be the maximum number of robots in a row or column of $grid_{\text{final}}$; and
 d be the maximum number of rows or columns of $grid_{\text{final}}$.
Calculate bounds $Y_{MAX} \leftarrow \text{FindYMAX}(n)$; $X_{MIN} \leftarrow \text{FindXMIN}(n)$;
repeat
 foreach $robot\ r_i \in R$ **do**
 Let j be the row index of r_i ($j \leftarrow Y_{MAX} - r_i.y + 1$);
 Priority of movement: Upper > West > East > South > North;
 // Case $\Psi 1$ (Robot in even row within bounds)
 if $j \leq d$ **and** $j \pmod{2} = 0$ **then**
 if $T_{x,y+1}$ **is empty** **then**
 r_i moves to $T_{x,y+1}$;
 $j \leftarrow j - 1$;
 else if $T_{x+1,y}$ **is empty** **then**
 r_i moves to $T_{x+1,y}$;
 // Case $\Psi 2$ (Robot beyond distribution depth)
 else if $j > d$ **then**
 if $T_{x,y+1}$ **is empty** **then**
 r_i moves to $T_{x,y+1}$;
 $j \leftarrow j - 1$;
 else if $T_{x+1,y}$ **is empty** **then**
 r_i moves to $T_{x+1,y}$;
 // Case $\Psi 3$ (Odd row alignment)
 if $j \pmod{2} = 1$ **and** r_i **not aligned correctly** **then**
 if r_i **is near** X_{MIN} **and** $T_{x-1,y}$ **empty** **then**
 r_i moves to $T_{x-1,y}$;
 else if $T_{x-1,y}$ **and** $T_{x-2,y}$ **empty** **and** **West move capable** **then**
 r_i moves to $T_{x-1,y}$;
 else if $T_{x+1,y}$ **and** $T_{x+2,y}$ **empty** **then**
 r_i moves to $T_{x+1,y}$;
 until $\text{IsUniformlyDistributed}(R, rc, d, Y_{MAX}, X_{MIN})$ **is True**;

Algorithm 3: FormGrid(R) — Part II

Input : Continuation of FormGrid(R)
Output: Robots move to form a uniform grid distribution
foreach $robot\ r_i \in R$ **do**
 Let j be the row index of r_i ($j \leftarrow Y_{MAX} - r_i.y + 1$);
 // Case $\Psi 4$ (Filling North rows)
 if $j \pmod{2} = 1$ **and** **other robots on even-indexed rows** **and** $\text{CheckXGap}()$ **then**
 if $r_i \in \text{first } rc \text{ robots}$ **and** $T_{x,y+2}$ **empty** **then**
 r_i moves to $T_{x,y+2}$;
 // Case $\Psi 5$ (Handling excess robots)
 if $j \pmod{2} = 1$ **and** $\text{CheckXGap}()$ **and** **no of robots in west** $\geq rc$ **then**
 if $j = 1$ **then**
 if $T_{x,y-2}$ **empty** **then**
 r_i moves to $T_{x,y-2}$;
 else if **West robots** $< rc$ **and** $T_{x-1,y}$ **and** $T_{x-2,y}$ **empty** **then**
 r_i moves to $T_{x-1,y}$;
 else
 if **other robots above odd rows** $< rc$ **and** $T_{x,y+2}$ **empty** **then**
 r_i moves to $T_{x,y+2}$;
 else if **no of robots in west** $< rc$ **and** $T_{x-1,y}$ **and** $T_{x-2,y}$ **empty** **then**
 r_i moves to $T_{x-1,y}$;
 else if $T_{x,y-2}$ **empty** **then**
 r_i moves to $T_{x,y-2}$;
 return History of movements for further use;

Fig: 1

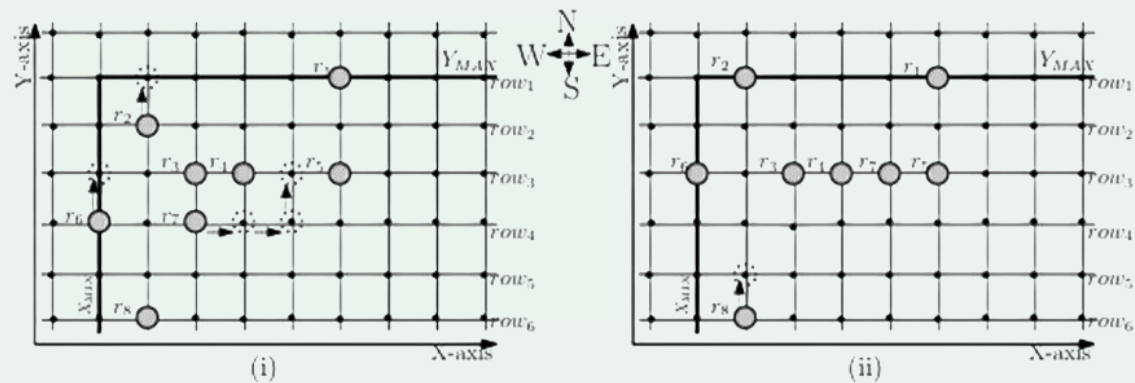


Fig: 2

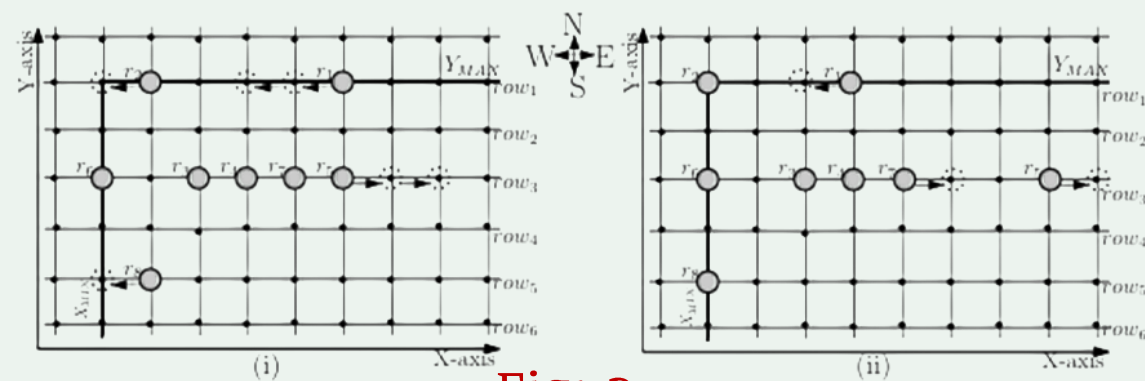


Fig: 3

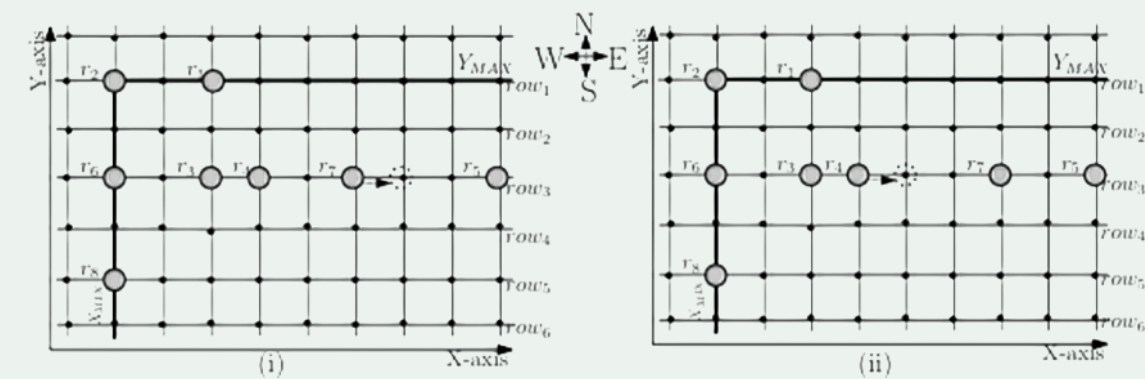


Fig: 4

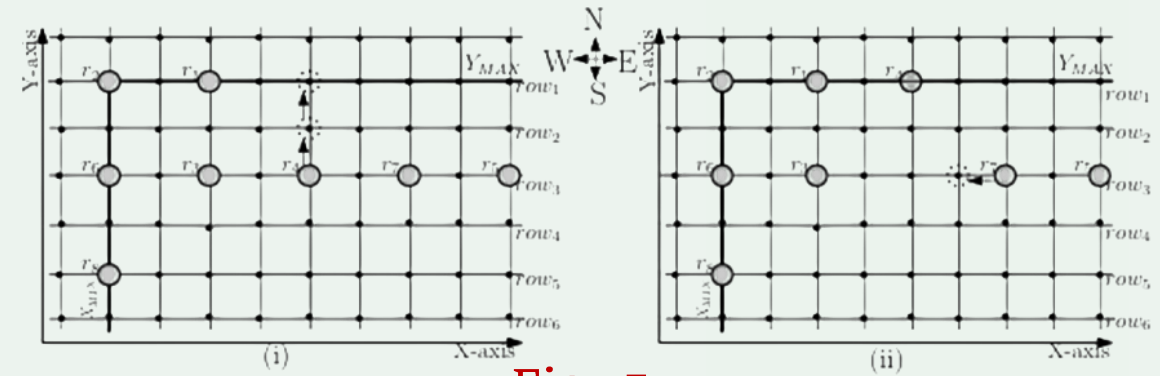


Fig: 5

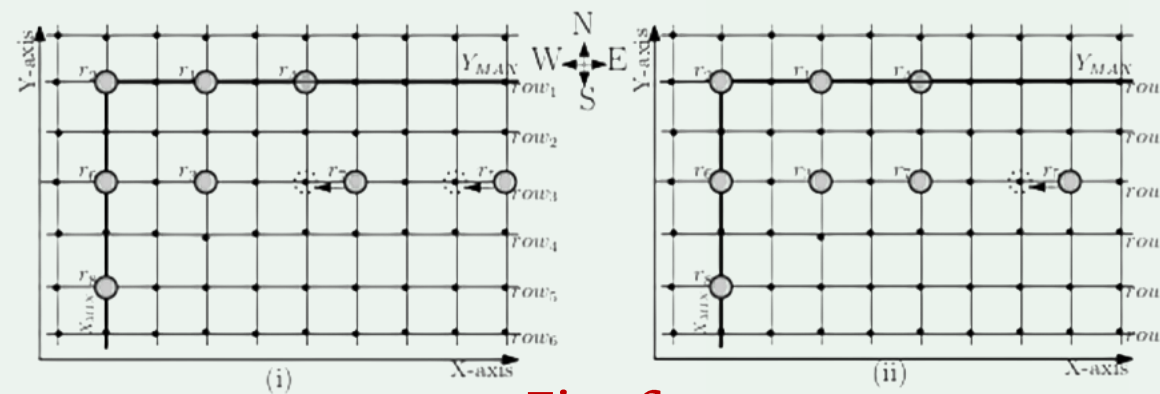
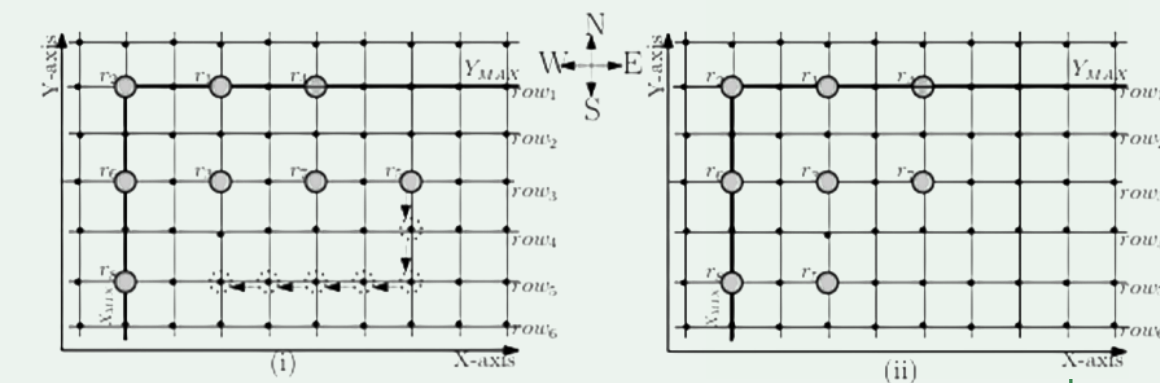
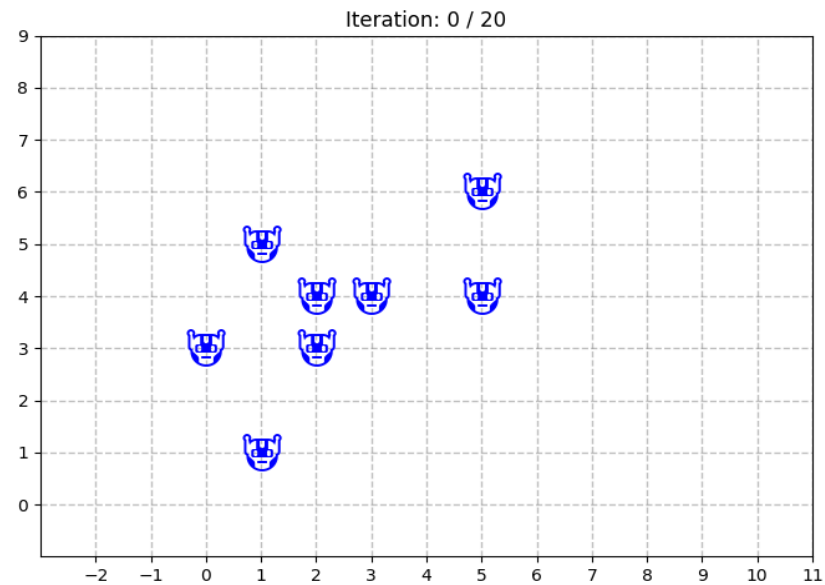


Fig: 6



OUTPUT

MAIN CODE LINK



Previous

Play

Next

Saving...

USE CASES

Military and Border Patrolling

Autonomous robots can be deployed for **patrolling sensitive regions** with minimal human presence.

Disaster Response Operations

Supports systematic and collision-free robot deployment in unknown terrains.



Surveillance & Area Monitoring

Ensures complete and non-overlapping coverage of the target area.

Agricultural Robotics

Provides equal spacing for planting, spraying, and crop monitoring.

FUTURE SCOPE

Asynchronous Execution Model

Adapt the solution for a **fully asynchronous (ASYNC)** environment, making it more practical for real-world deployment.

Fault Tolerance

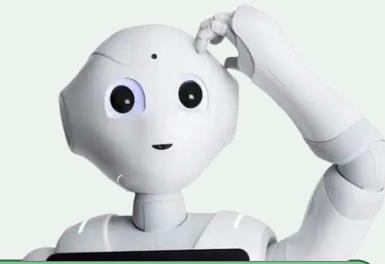
Modify the algorithm to handle **robot failures or malfunctioning robots** without affecting the final grid formation.

Obstacle-Aware Scattering

Extend the model to support **obstacles or irregular terrains** within the grid environment.

Dynamic Environment Handling

Enable robots to **re-scatter automatically** when new robots are added or existing robots are removed.

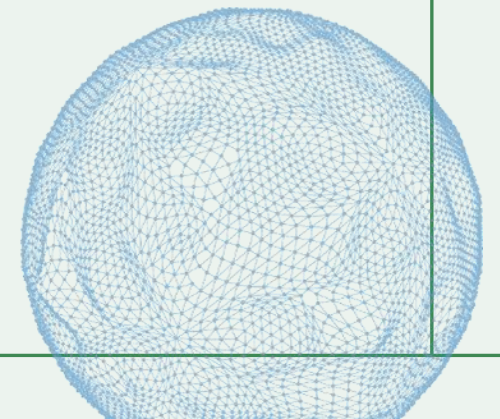
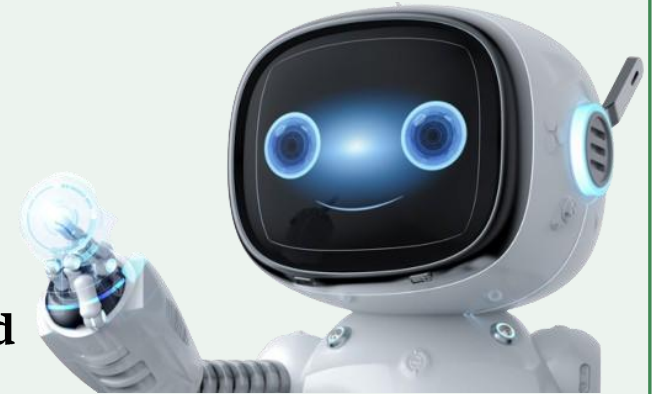


CONCLUSION

In this presentation, we explored a distributed approach for the **uniform scattering of autonomous robots on alternate nodes of a two-dimensional grid**.

By using **simple local observations** and operating under a **semi-synchronous model**, the proposed algorithm successfully achieves **collision-free, deadlock-free, and finite-time convergence** without any communication or memory.

This work demonstrates how **local decision-making by identical and oblivious robots** can lead to a **well-organized global structure**, highlighting the strength and scalability of swarm robotic systems. Such approaches are highly applicable in **area coverage, surveillance, and distributed sensing**, especially in large and unknown environments.



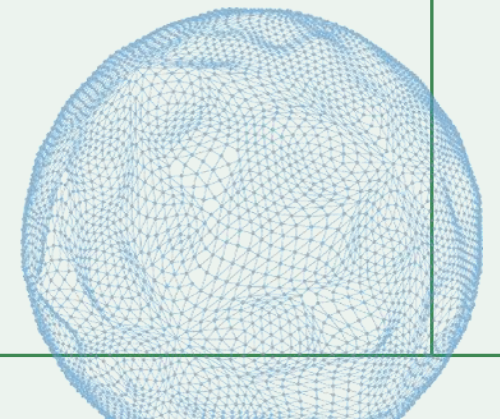
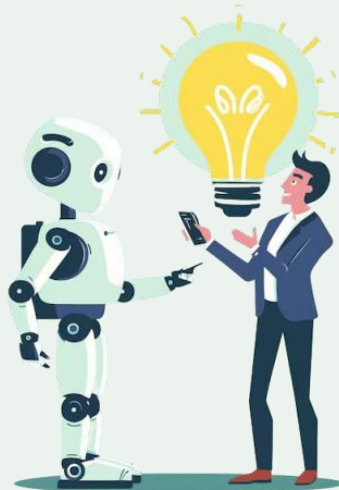
OUR LEARNING

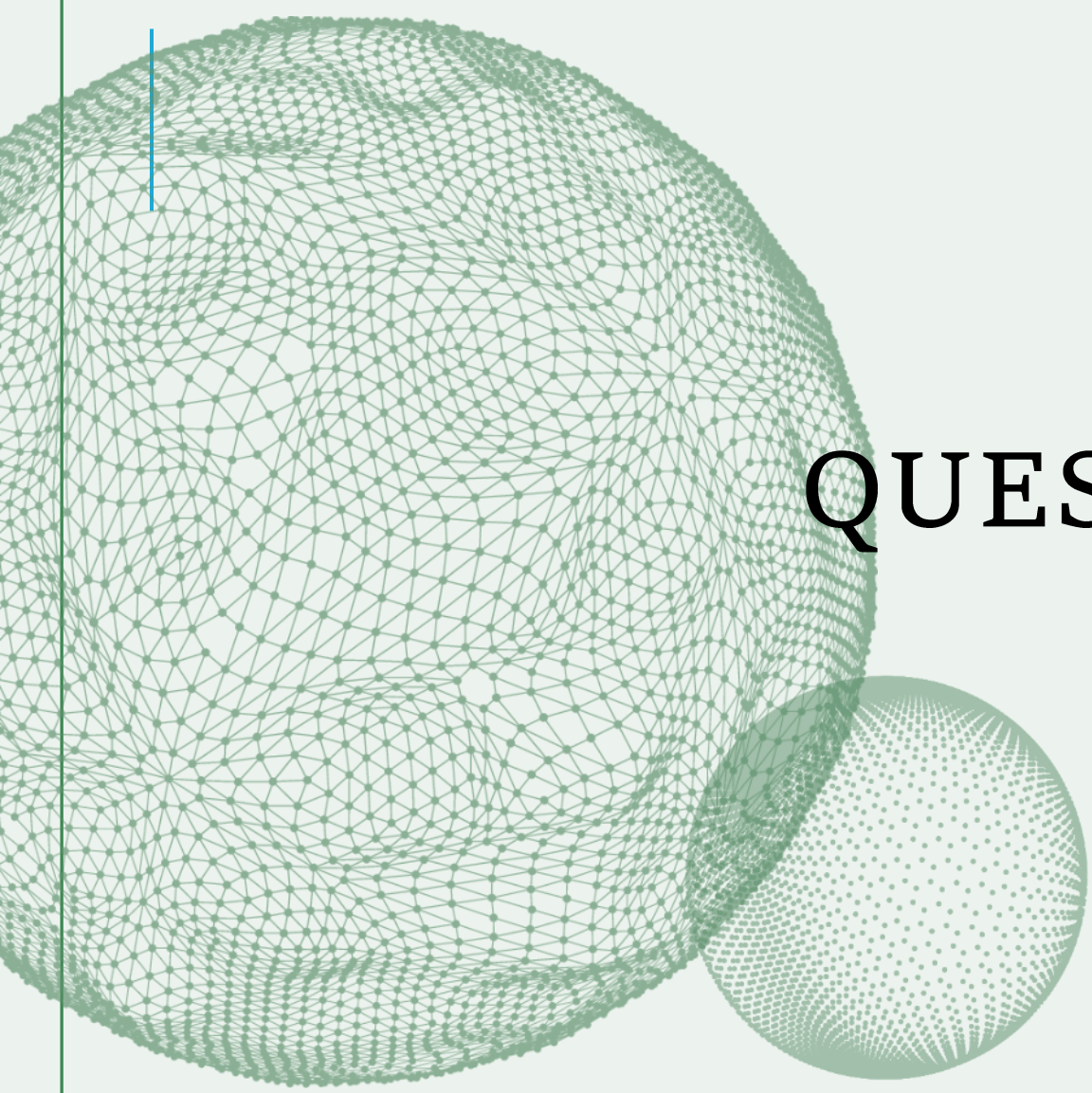
- Gained a clear understanding of **swarm robotics and distributed robot coordination**.
- Learned how **simple local rules** can produce **organized global behavior** without centralized control.
- Understood the **Look-Compute-Move cycle** and **semi-synchronous execution model**.
- Analyzed how **collision-free and deadlock-free algorithms** are designed.
- Improved knowledge of **grid-based movement and uniform scattering problems**.
- Enhanced skills in **reading and interpreting research papers**.
- Developed **team collaboration and presentation skills** through group work.



REFERENCES

1. M. Mondal, S. G. Chaudhuri, and P. Chatterjee,
“Uniform Scattering of Robots on Alternate Nodes of a Grid,”
Department of Information Technology, Jadavpur University, Kolkata, India.
2. M. Dorigo and E. Şahin,
“Swarm Robotics – A Review,” Springer, 2004.
3. S. Das, P. Flocchini, G. Prencipe, *“Uniform Scattering of Robots on a Grid,”*
Proceedings of the International Conference on Distributed Computing Systems, IEEE.





QUESTIONS?

An abstract graphic on the left side of the slide features two overlapping spheres. The larger sphere in the background is constructed from a dense network of green dots connected by thin lines, creating a wireframe effect. A smaller, similar sphere is positioned in front of it to the right. A thin vertical blue line is visible on the far left edge of the larger sphere.

THANK YOU FOR LISTENING